不合拍的风：塔科马大桥的陨落

The Wind That Kept No Time: The Fall of Galloping Gertie



风没按节拍吹，大桥为何还是倒了？

READ 纯净阅读 | CHECK 中文核对 | LEARN 词汇记忆 | RECALL 回想巩固

英文原文

On the morning of 7 November 1940, a sustained wind of about 40 miles an hour set the Tacoma Narrows Bridge twisting. Its deck tilted from side to side in ever more violent oscillations until most of the main span tore apart and tumbled into the water. No one died. Footage of the spectacle would later fascinate students worldwide.

The bridge had opened in July 1940 as the third-longest suspension span in the world at the time, its roadway hung from steel cables. Girders only eight feet deep stiffened the deck, making it slender, elegant, and economical, but far from rigid. Even faint breezes made it heave, earning the nickname Galloping Gertie.

Textbooks have long invoked resonance: a periodic push that matches an object's natural frequency can amplify its motion, as when well-timed shoves propel a child on a swing. The analogy is intuitive but deceptive. That morning's wind did not pulse in time with the bridge; it blew fairly steadily.

The mechanism was subtler. Each twist altered the passing airflow, which reinforced the next twist: a feedback loop that kept tapping the wind for energy. In March 1941 a federal panel blamed the deck's flexibility and wing-like lift. Later analysis called the mechanism flutter; in February 1991 two engineers asserted in a physics journal that textbooks had erred.

The panel urged wind-tunnel tests, and trials on scaled-down replicas gradually became customary for suspension bridges. A sturdier replacement opened in October 1950. Gertie's fall may still instruct engineers, if read accurately: the wind never kept time with the bridge, the bridge's own motion turned a sustained wind destructive.

中文译文

1940年11月7日上午，时速约40英里的持续大风使塔科马海峡大桥开始扭转。桥面左右倾斜，振荡越来越剧烈，最终主跨的大部分断裂，坠入水中。无人遇难。这场景象的影像，后来吸引了世界各地的学生。

大桥于1940年7月通车，其悬索桥跨径当时位居世界第三，路面由钢缆悬吊。桥面只靠高八英尺的钢梁加强，因此显得纤细、优美又经济，却远不够刚硬。即使微风也会让它起伏，因而得了“奔腾的格蒂”这个绰号。

教科书长期用共振来解释这起事故：如果周期性的推力与物体的固有频率相合，就能放大运动，就像踩准节拍推秋千上的孩子。这个类比直观，却会误导人。那天上午的风并没有跟着大桥的节拍一阵阵吹来，而是相当平稳。

实际机制更微妙。桥面每一次扭转都会改变掠过的气流，气流又加强下一次扭转，形成不断从风中获取能量的反馈回路。1941年3月，一个联邦调查委员会把事故归因于桥面的柔性和类似机翼的升力。后来的分析把这一机制称为颤振；1991年2月，两位工程师在一份物理学期刊上指出，教科书的解释错了。

委员会力主做风洞试验，此后用缩尺模型测试悬索桥逐渐成为惯例。更坚固的新桥于1950年10月通车。只要读懂真正的机制，格蒂的倒塌仍能给工程师以启示：风从未与桥合拍，是桥自身的运动，让一股持续的风变得具有破坏力。

核心词汇

先记这22项。连着下方的原文短语一起回想。

sustained adj.
持续的
a sustained wind of about 40 miles an

oscillation n.
振荡
side in ever more violent oscillations

tear apart phr.v.
撕裂；断裂
the main span tore apart and tumbled

footage n.
影片片段；影像
Footage of the spectacle would later

suspension n.
悬吊；悬索结构
as the third-longest suspension span

span n.
桥跨；跨径
as the third-longest suspension span

rigid adj.
刚硬的；不易变形的
and economical, but far from rigid.

heave v.
上下起伏
Even faint breezes made it heave,

invoke v.
援引；用作解释
Textbooks have long invoked resonance:

resonance n.
共振
Textbooks have long invoked resonance:

periodic adj.
周期性的
periodic push that matches an object's

frequency n.
频率
object's natural frequency can amplify

amplify v.
放大；增强
object's natural frequency can amplify

analogy n.
类比
The analogy is intuitive but

intuitive adj.
直观易懂的
analogy is intuitive but deceptive.

deceptive adj.
具有误导性的
analogy is intuitive but deceptive.

mechanism n.
机制；机理
The mechanism was subtler.

feedback n.
反馈
a feedback loop that kept tapping the

flexibility n.
柔性；易弯曲性
panel blamed the deck's flexibility

scaled-down adj.
按比例缩小的
scaled-down replicas gradually became

customary adj.
惯常的
replicas gradually became customary

destructive adj.
具有破坏力的
turned a sustained wind destructive.

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

twist v.
扭转

deck n.
桥面

tilt v.
倾斜

violent adj.
剧烈的

tumble v.
翻落；坠落

spectacle n.
壮观的景象

fascinate v.
使着迷；吸引

hang from phr.v.
悬挂于

girder n.
大梁；钢梁

stiffen v.
使更坚硬；加固

slender adj.
纤细的

elegant adj.
优美的

economical adj.
经济实惠的

faint adj.
微弱的

breeze n.
微风

nickname n.
绰号

textbook n.
教科书

well-timed adj.
时机恰当的

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

shove n.
推；猛推

propel v.
推动

pulse v.
有节奏地起伏

steadily adv.
平稳持续地

subtle adj.
更微妙的

alter v.
改变

reinforce v.
加强

tap v.
汲取；利用

panel n.
专家小组；调查委员会

assert v.
断言；明确提出

err v.
犯错

wind-tunnel adj.
风洞的

replica n.
复制模型

gradually adv.
逐渐地

sturdy adj.
更坚固的

replacement n.
替代物；此处指新桥

instruct v.
指导；给以启示

accurately adv.
准确地

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 a _____ wind of about 40 miles an hour

02 can _____ its motion,

03 a _____ loop that kept tapping the wind for energy.

04 gradually became _____ for suspension bridges.

文章主线

- 事故 1940年，刚通车数月的塔科马大桥在持续大风中剧烈扭转、断裂坍塌，无人遇难。
- 隐患 轻巧经济的桥面刚度不足，连微风都能使其起伏，早已有“奔腾的格蒂”之名。
- 纠误 课本常用共振解释事故，但当天的风相当平稳，并没有按桥的节拍反复推它。
- 机制 桥的扭转改变气流，气流又加强下一次扭转，不断从风中取能；这一机制称为颤振。
- 后续 事故推动悬索桥模型风洞试验成为惯例；更坚固的新桥于1950年通车。

参考答案　1 sustained 2 amplify 3 feedback 4 customary